Grid Stress Monitoring, Demand, & Growth Modeling
DS591 Final Project Executive Report

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Modern electrical demands are evolving as industrial growth and compute-intensive infrastructure expand. Electrical grids must balance need with margins to avoid overstress on the system. With limited visibility into real-time grid stress signals or long-term trends, it's hard for operators to make these balancing decisions quickly. Our project proposes a solution by creating a pipeline that monitors real-time electrical grid stress in New England and contextualizes this with multi-year historical consumption data. This system will enable predictive demand modeling and real-time anomaly detection to provide grid operators actionable insight. We aim to predict seasonal demand, weekly operational patterns, impact of new AI infrastructure, and forecast accuracy degradation. These predictions will be made through identifying volatility, peak stressor events, and overall growth in industrial and commercial electrical usage.

The data for our pipeline is from two sources ISO-NE and EIA. ISO-NE oversees operation of New England's bulk electric power system, and EIA or the US Energy Information Administration has historical data on electric usage by state and region. We use ISO-NE's streaming and historical data and EIA's historical data for this project. The core concepts of this data revolve around electricity consumption and grid load variables. `total_load_mw` is the total amount of electricity currently being consumed in megawatts (MW). `target_load_mw` is the next 5-minute interval load used as our model target. The datasets have rolling averages over a range of intervals, such as the last 3 and 12 intervals. There is also a percentage change in demand from the previous interval and delta of demand which is in raw megawatts. There is a binary `stress_flag` that marks when the grid is under stress. Stress is defined as if the load to capacity ratio is over 85%. We also have access to sales data in units of kilowatt-hours (kWh) such as electricity sold in a sector per month, sales per customer, and YOY growth.

Through exploratory data analysis, we observed in total_load and total_rolling averages that the rolling average flattens out any spikes. In the first day of collection, the morning had a dip to 8500 MW and evening spike of 12500 MW, and the rolling average followed in direction hovering between 10400 and 10600 MW. This demonstrates rolling average captures the trend direction. In the EIA data we observed sales by sector show industrial usage stays low compared to residential and commercial usage that oscillates between 3000 to 5000 million kWh, demonstrating the ability to observe historical patterns from our data. From this and further analysis in our codebase, we were able to spot several identifiable patterns and key insights. We found that seasonal load variation is between about 17000 MW in the summer to about 10000 MW in the spring. Also between states there were distinct patterns, such as RI and MA with major cooling usage where ME and VT have high heating usage comparatively to other states in New England. A key insight we observed was that all 6 New England states (CT, MA, ME, RI, NH, VT) show a negative compounded annual growth rate and a general decline in total industrial load.

After this exploration, we developed our model and evaluation. Our target is to predict the next five minute interval to total system load, or what the load will be 5 minutes from current time. We attempted four different variations of modeling to see what would fit best. We performed baseline, linear regression, random forest, and XGBoost. To get the best fit of

parameters, we ran nine combinations for XGBoost and random forest, and time series split of three. To prevent data leakage, rolling averages are computed as backward-looking windows only, ensuring no future data is used in predictions. The results from our models showed to be pretty close to the actual occurrence. To confirm our results as statistically significant, we checked MAPE, RMSE, and F1. MAPE is the mean absolute percentage error to statistically measure accuracy, and RMSE is the root mean square error to measure distance between actual and estimated values. We also wanted to check that models weren't overfitting to the data since the results were very accurate. Linear regression performed the best at 0.42% MAPE and wasn't flagged as being overfit. Random forest 2.22x and XGBoost 2.39x flagged as overfit and we believe this is because the models memorized autocorrelation patterns from the summer/fall/winter training period. Our model results are significant in that they perform better than ISO-NE's prediction percentage. Interestingly, random forest had a better precision in stress detection meaning fewer false alarms. With these tests and validations we ran, we were able to confirm that the observations made were significant. All technical implementation can be observed in more detail in our github repository.

Our initial predictions of what the data would look like didn't exactly match reality. We were able to confirm seasonal patterns and forecast accuracy degradation over time, but our thoughts on AI infrastructure and weekly operational patterns didn't match the results. The ability to clearly see stress indicators, historical consumption patterns, and models for future stress events is really beneficial in effectively managing electrical grid usage. While one limitation of our pipeline is that our streaming data is microbatched so limiting functionality, the statistical results are still impactful for our timeframe. One major challenge was difficulty with RBAC and user permissions. A specific example is with Synapse Serverless SQL, the credential uses a SAS token stored directly in the database. We wanted to store that credential in a secret key, but due to Azure Key Vault access restrictions in the BU-managed resource group that could not be resolved within the timeframe. In a production environment, this secret would be stored in Azure Key Vault and referenced via a Key Vault-backed managed identity. We refer to this specific instance as it is a known data governance concern we would rectify in future work. An interesting avenue of future exploration would be to measure regions with confirmed data center development and see what patterns unfold there. Additionally, we replayed ISO-NE demand data through Azure Event Hub and Stream Analytics into a live Power BI dashboard. Cadence was accelerated from the standard 5-minute interval to near-instant, solely for demonstration purposes, with the dashboard visibly updating in real time as each event was processed. This project was very insightful to show that dynamic electric grid stress monitoring, demand, and growth modeling is feasible. In our modern age of technology, this could be very useful in real-world applications and a great area for further development and research.